PROTOCOLS DTP, VTP,CDP AND LLD

DTP

1. Description: **DTP, Dynamic Trunking Protocol,** is a trunking protocol that is developed and proprietary to Cisco which is used to automatically negotiate trunks between Cisco switches. Trunk negotiations are managed by DTP only if the port is directly connected to each other.

2. Details:

The DTP, Dynamic Trunking Protocol, '**switchport mode'** interface subcommand is used to define the administrative trunking mode that tells the interface whether to trunk always or not or negotiate to trunk or not.

- **`switchport mode access`**

Port works only as access, allows just one VLAN. Never trunks, even if the other side is trunk.

- **`switchport mode trunk`**

Forces trunking. Tries to make the other side trunk too, but will still trunk even if the neighbor doesn't.

- **`switchport mode dynamic auto`**

Waits for the other side to start trunking. Becomes trunk only if the neighbor is set to trunk or dynamic desirable.
(Default on newer switches like 2960, 3560)

- **`switchport mode dynamic desirable`**

Actively tries to trunk. Becomes trunk if the other side is set to trunk, dynamic desirable, or dynamic auto.
(Default on older switches like 2950, 3550)

- **`switchport nonegotiate`**

Disables DTP completely. Used when you manually configure trunk and don't want any negotiation.

The table below describes how an interface selects its trunking operational mode based on its own trunking administrative mode and the results of the DTP process.

Trunk Mode	Dynamic Auto	Dynamic Desirable	Trunk	Access
Dynamic Auto	Access	Trunk	Trunk	Access
Dynamic Desirable	Trunk	Trunk	Trunk	Access
Trunk	Trunk	Trunk	Trunk	Limited connectivity
Access	Access	Access	Limited connectivity	Access

3. Recommandations:

- **Disable DTP on Access Ports**

→ Prevent trunking:
```
switchport mode access
switchport nonegotiate
```

- **Use Manual Trunk Configuration**

→ Set trunks explicitly:
```
switchport mode trunk
switchport nonegotiate
```

- **Avoid Dynamic Modes**

→ Don't use `dynamic auto` or `dynamic desirable`

- **Restrict Allowed VLANs on Trunks**

→ Limit VLANs:
```
switchport trunk allowed vlan 10,20
```

- **Audit Trunk Interfaces**

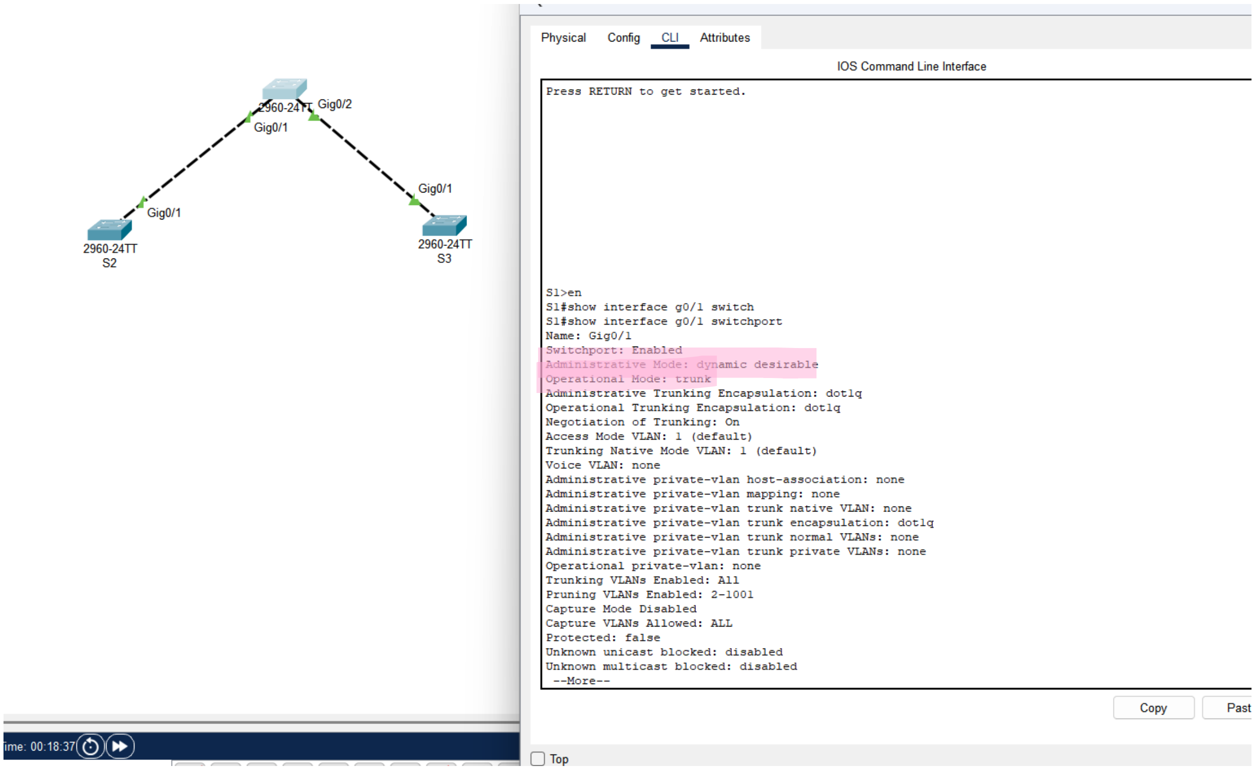
→ Check with:
```
show interfaces trunk
```

4. References:

https://study-ccna.com/dynamic-trunking-protocol-dtp-cisco/

https://itexamanswers.net/3-5-5-packet-tracer-configure-dtp-instructions-answer.html

5. Affected Locations – Main Switch



VTP

Description: **VTP (VLAN Trunking Protocol)** is a Cisco proprietary protocol used by Cisco switches to exchange VLAN information. With VTP, you can synchronize VLAN information (such as VLAN ID or VLAN name) with switches inside the same VTP domain. A VTP domain is a set of trunked switches with the matching VTP settings (the domain name, password and VTP version). All switches inside the same VTP domain share their VLAN information with each other.

Details:

To better understand the true value of VTP, consider an example network with 100 switches. Without VTP, if you want to create a VLAN on each switch, you would have to manually enter VLAN configuration commands on every switch! VTP enables you to create the VLAN only on a single switch.

VTP modes:

- **Client Mode**

Can’t create or delete VLANs. Accepts and forwards VLAN updates from servers.

- **Server Mode**

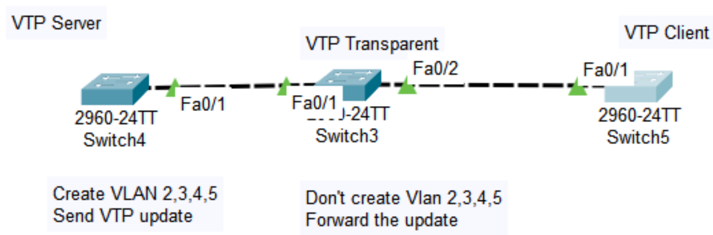
Can create, delete, and send VLAN updates to other switches. *(Default mode)*

- **Transparent Mode**

Can create and delete VLANs locally, but doesn't share them. Only forwards updates from others.

- **Off Mode**

Like transparent, but doesn't even forward updates. Only in VTP version 3.



Recommandations:

- **Server Mode** – Use on main/core switch. Manages VLANs. Secure it with a password.
- **Client Mode** – Use on access switches. Gets VLAN updates but can't make changes.
- **Transparent Mode** – Use if you want to manage VLANs locally without sharing them. Still forwards updates.
- **Off Mode** – For full isolation from VTP (VTPv3 only). Doesn't send or forward anything.

References:

https://study-ccna.com/vtp-modes/

https://www.cisco.com/c/en/us/support/docs/lan-switching/vtp/98154-conf-vlan.html

CDP

Description: Cisco Discovery Protocol (CDP) is an OSI Layer 2 protocol that operates between Cisco devices such as routers and switches. CDP messages contain information about the device such as device

ID, platform, connected interface, IOS version, and Layer 3 address. Because CDP operates at Layer 2, only directly connected devices exchange informatio

Details: CDPis enabled on the system and on the interfaces by default. If you prefer not to use the CDPdevice discovery capability, you can disable it with the no cdp run command at the system level and no cdp enable command at the interface level.

Recommandations:

- **Disable CDP on Unused or Untrusted Ports**

Prevents leaking device info to unauthorized devices.

- **Keep CDP Enabled on Trusted Switch-to-Switch Links**

Helps with network mapping and troubleshooting.

- **Avoid Using CDP on Internet-Facing Interfaces**

Exposes details like device type, software version, and IP—use `no cdp enable`.

- **Use CDP for Troubleshooting and Documentation**

It's useful for discovering neighbor devices, especially in large networks.

References: https://learningnetwork.cisco.com/s/article/cisco-discovery-protocol-cdp-x

Description: Link Layer Discovery Protocol (LLDP) is a layer 2 neighbor discovery protocol that allows devices to advertise device information to their directly connected peers/neighbors. It is best practice to enable LLDP globally to standardize network topology across all devices if you have a multi-vendor network.

Details:

Commonly used layer 2 discovery protocols are often vendor-proprietary, for instance, Cisco's CDP, Foundry's FDP, Extreme's EDP and Nortel's NDP. This makes layer 2 discovery difficult in a heterogeneous environment. To counter this, IETF has introduced a standard vendor-neutral configuration exchange protocol – the LLDP.

Using LLDP, device information such as chassis identification, port ID, port description, system name and description, device capability (as router, switch, hub…), IP/MAC address, etc., are transmitted to the neighboring devices. This information is also stored in local Management Information Databases

(MIBs), and can be queried with the Simple Network Management Protocol (SNMP). The LLDP-enabled devices have an LLDP agent installed in them, which sends out advertisements from all physical interfaces either periodically or as changes occur.

Recommandation:

- **Enable LLDP on Device Uplinks and Switch-to-Switch Connections**
  Useful for network discovery, mapping, and troubleshooting—especially in mixed-vendor environments.
- **Disable LLDP on Unused or Untrusted Ports**
  Prevents leaking device information to unauthorized users.
- **Don’t Use LLDP on Internet-Facing Interfaces**
  Avoid exposing device details (hostname, port, IP, OS, etc.) externally.
- **Use LLDP with VoIP Phones and Access Points**
  Helps deliver correct VLAN and power settings via LLDP-MED (Media Endpoint Discovery).

References: https://www.whatsupgold.com/network-discovery/link-layer-discovery-protocol-lldp